PS000109 Synopsis – Exposure-Triggered Worker Health Screening (Kachchh)

Oct 31, 2026 · @Neel and Amee

The pitch

For Kachchh's port, logistics, cement and salt-pan workers, we measure **heat and dust** at the worksite and on the worker, use that to trigger the **right check-up at the right time**, and keep a health record the **worker owns**, so follow-up does not stop when they change employer.

Problem ID	PS000109
Statement	Digital Occupational Health Monitoring for Industrial Workers
Posted by	District Development Office, Kachchh (Healthtech)
Users named	Labourers, localities
Expected outcome (their words)	A secure digital system for periodic health screening, risk alerts and follow-up tracking
Synopsis deadline	6 Nov 2026 · shortlist 23 Nov · finale 27 Nov

The problem

Workers' exposure is never measured, so check-ups happen late or not at all, and records break every time a worker moves. The statement lists four failures: records kept separately, missed periodic screening, risks not tracked, and workers changing employers or locations.

Why it matters, in numbers:

- **Heat costs India about ₹4,500 crore a day.** The Lancet Countdown 2025 puts 2024 losses at $194 billion and 247 billion work hours (Outlook Business).
- **Heat is already past safe limits for this kind of work.** In Tamil Nadu salt pans, 95% of heat measurements exceeded the limit and 93% of 352 workers had heat-strain symptoms (Venugopal 2023, via PMC). Kachchh has the same salt-pan work and hotter summers; no Kachchh study exists yet.

- **Occupational disease is almost never reported.** Only 132 cases were notified, by just 5 states, across 2014–2016 (ILO statement).
- **Dust disease has no reliable count in Gujarat.** Down To Earth reports 182 silicosis cases and 80 deaths in Morbi ceramics since 2018, and says there are "no reliable estimates" (DTE).
- **Workers hide symptoms.** The Health Department's sister statements (PS000247, PS000271) say workers fear job loss and employers fear liability, and that dust exposure "is rarely measured at an individual level, so exposure and health records are not linked".

Narrowed scope

One district, two hazards, five job groups. We build for Kachchh only, measure heat and dust only, and cover the jobs where those two hazards are worst. Everything else is a later module.

Why heat and dust: they are Kachchh's two most widespread hazards, both are cheap to sense with parts we already use, and both map to checks with a clear medical basis (silicosis is a notifiable disease; heat damages kidneys).

In scope (build for the finale)	Out of scope (later, say so in the pitch)
Kachchh district	Other districts, statewide rollout
Heat (site sensor + forecast) and dust (personal badge)	Noise, chemicals, vibration
5 job groups: port yard and crane workers, truck drivers, cement plant workers, salt-pan workers (Agariyas), outdoor construction and loading	Office, skilled-trade and indoor jobs
Rules that turn exposure into "which check-up, when"	Machine-learning risk prediction (needs outcome data we don't have)
Worker record, alerts in Gujarati and Hindi, follow-up tracking, weekly symptom check-in by SMS or voice call	Heart rate from cheap bands, eye-photo checks, Bluetooth zone tags, X-ray AI
Anonymous district dashboard for the DDO	Employer-facing HR or compliance software

The job-to-test rulebook stays small (5 jobs × 2 hazards). It is necessary plumbing, not the innovation: on its own it is just a lookup table. The new parts are measuring each worker's real exposure, triggering screening from it, and a record the worker controls.

Next modules we will show, but not build, for the finale:

- **Heart rate from a cheap fitness band** (optional, volunteers only): heat around the worker plus heat inside the body gives a stronger "stop and rest" alert. We read it from a band, we don't build the sensor.
- **Eye checks linked to exposure:** a dust spike prompts an eye photo; salt-pan workers get a monthly photo for glare damage. QR card for identity, no face recognition.
- **Bluetooth zone tags (~₹300 a worker):** one sensor per work zone, so each worker's dose is estimated without a badge for everyone.
- **Noise:** the easiest next hazard (a ₹200 microphone on the same board).

Our solution: the core loop

The system measures exposure, decides which check-up is due, alerts and books it, then keeps the case open until a result is logged.

Measured exposure triggers the check-up, and follow-up closes the loop



core loop · 6 steps

Step 3 is the heart: each worker's measured heat and dust dose is compared with limits and their job's rules. The record (step 5) stays with the worker, so a move to a new employer does not reset it.

Where the intelligence is, and where the data comes from

We do not claim a trained health-risk model, because no labelled data on Kachchh workers' outcomes exists. The system is smart in three layers, and only one needs training, on data we collect ourselves.

1. **Heat-strain index: no training needed.** WBGT (wet-bulb globe temperature, the standard heat-stress measure) is calculated from temperature, humidity, sun and wind using published formulas, then matched to standard work–rest tables (ISO 7243 / ACGIH). We validate our sensor against a reference instrument.
2. **Exposure anomaly detection: trained on our own sensor data, no labels needed.** The model learns what "normal" dust and heat look like for each site and job, then flags spikes and slow drifts. This is the same idea as our Dr.Panels residual detector.
3. **Risk prediction: future, needs partners.** Predicting who will get sick needs real outcomes (test results over time). Until a partner shares de-identified records, this stays on the roadmap, clearly labelled.

Data	Source	Used for	Status
Weather and forecasts	Open-Meteo, IMD	Heat alerts a day ahead	Public, free
Site heat and personal dust readings	Our sensors at a pilot site	Exposure dose, anomaly model	We collect during build
Exposure limits and check-up rules	OSH Code schedules, ISO 7243 / ACGIH, Indian studies (see JEM sheet)	Rulebook	Public
Worker records	Our app, with the worker's consent	Scheduling, follow-up	Generated in pilot
Health outcomes (labels)	NIOH Ahmedabad, ESIC dispensaries, factory health centres, PTRC case records	Future risk model	Needs partnership

Rule for the whole team: no made-up data presented as real. Demo data is either measured by our sensors or clearly marked as sample.

Weekly symptom check-ins (headache, dizziness, cough, dark urine) are how outcome data starts to exist: over months, exposure plus symptoms becomes the labelled data a future risk model needs.

Where to download the heat and dust data now: Open-Meteo historical weather and Open-Meteo air quality (hourly, CSV), NASA POWER as a backup, IMD and CPCB for official records. Points: Bhuj 23.25, 69.67 · Kandla 23.03, 70.22 · Mundra 22.74, 69.70 · Little Rann 23.35, 70.70 · Abdasa 23.43, 68.85 (approximate).

What already exists, and how we differ

Each piece exists separately; nobody we found connects measured exposure to triggered screening and a record that follows the worker.

Product	What it does	What it misses
OccuCare (Ahmedabad)	Employee health records, periodic exams, compliance forms	No exposure data coming in; one employer per record
Jarsh Safety, Proxgy SmartHat	Smart helmets: GPS, SOS, falls, voltage, cooling	Accidents and emergencies, not long-term health
Blackline, Honeywell, MSA, Dräger	Connected gas detectors	Expensive; gas only; no health follow-up
Lab dust samplers, noise dosimeters	Accurate audit measurements	A few days a year; results never reach the worker's record
Consumer fitness bands	Heart rate, steps	No exposure; data stays in a personal app

Our one-line answer to judges: "OccuCare stores check-up results. We decide when a worker needs a check-up, from what they were actually exposed to, and keep that record with the worker."

Legal basis

The law already requires most of what we build; factories just lack an easy way to do it.

Section 41C, Factories Act 1948 (occupier's duties in hazardous-process factories), summarised from the bare act:

- **(a) Health records:** keep accurate, up-to-date health records of workers exposed to harmful substances, and let workers access them. This supports a worker-owned record.
- **(b) Competent supervision:** appoint qualified people to supervise hazardous substances and provide protective facilities.
- **(c) Medical exams:** examine every worker before they start the job, while in it, and after they leave it, at intervals not exceeding 12 months. The "after they leave" part is exactly the follow-up that breaks today.

OSH Code 2020, in force from 21 Nov 2025. It consolidates 13 labour laws including the Factories Act, and makes employers provide annual health check-ups for prescribed employees (Taxmann). Its Third Schedule lists 29 notifiable diseases, including silicosis (#12) and noise-induced hearing loss (#22) (list). We cite the OSH Code first and the Factories Act as background.

Gujarat Factories (OHS Audit) Rules, 2025, dated 1 March 2025 and published around July 2025 (Lexplosion):

Category	Which factories	External audit every
I	Major accident hazard installations	2 years
II	Hazardous processes, or hazardous chemical storage above 100 kg/L	3 years
III	All other factories licensed for 100+ workers	4 years

All three also need an internal audit every year, unless an external audit happens that year. The summary we read does not say whether audits check medical surveillance; if they do, our records are ready-made audit evidence. Open question for the team: read the Rules' full text.

Tech stack and parts

Two devices and one web app. Prices are approximate retail estimates, not quotes.

Part	Role	Approx. cost
ESP32 dev board	Brain + Wi-Fi/Bluetooth for both devices	₹400 (we own some)
BME280	Air temperature, humidity, pressure (site node)	₹300 (owned)
DS18B20 inside a black-painted ball	Black-globe temperature for WBGT (sun heat)	₹150
PMS5003 or similar	Dust (PM2.5 / PM10) on the personal badge	₹800–1,200
Battery, charger, enclosure	Field use	₹300–500

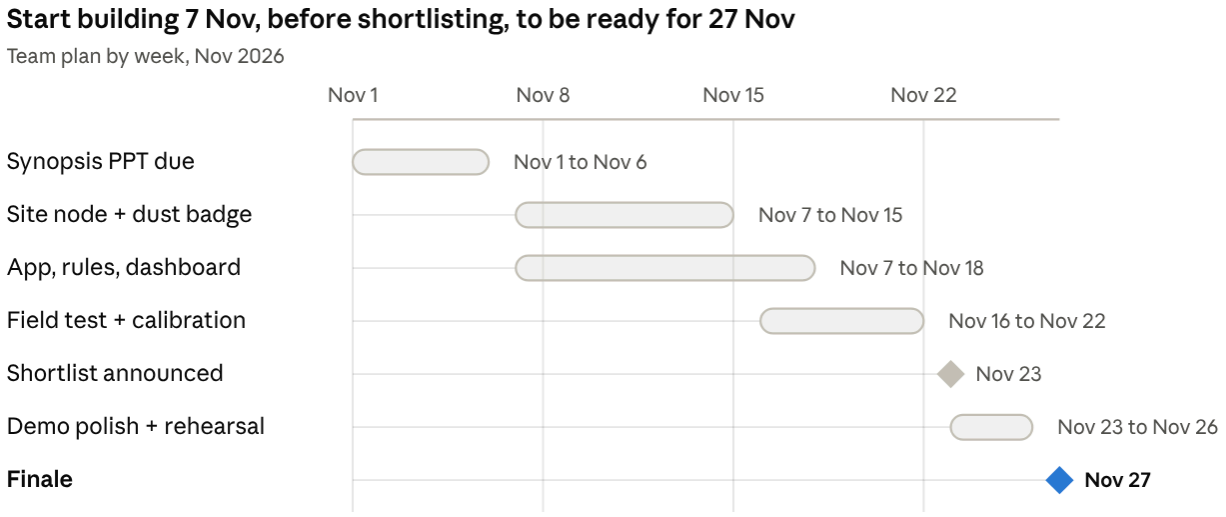
About ₹2,000 per site node and ₹1,500 per badge.

Software: a FastAPI backend with a SQL database; a phone-first web app in Gujarati, Hindi and English (QR worker card, check-up schedule, follow-up); SMS or voice-call alerts for workers without smartphones; a DDO dashboard showing only anonymous totals.

A cheap dust sensor cannot tell silica from other dust, so the badge triggers a dust check-up and the silica rule comes from the job type. We say this openly.

Build plan and roles

The synopsis is due 6 Nov, and only 4 days separate the shortlist (23 Nov) from the finale (27 Nov), so we build from 7 Nov without waiting.



build plan · Nov 2026

Work split by role, not by name:

Role	Owns
Hardware	Site node, dust badge, calibration
Backend	API, database, rules engine, alerts
App and UI	Worker app, Gujarati/Hindi text, DDO dashboard
Data and ML	WBGT calculation, anomaly model, demo data checks
Field and research	Pilot site, legal and medical rules, expert review
Pitch	Synopsis PPT, demo script, judge Q&A

First steps this week:

- [ ] Email the DDO resource person (parshuuniyal2019@gmail.com) with our five questions
- [ ] Agree scope and assign roles on a team call
- [ ] Order parts today: 2 × PMS5003, 2 × DS18B20 + black ball, extra ESP32 and BME280, batteries, enclosures
- [ ] Download the Kachchh heat and dust CSVs (links in the data section)
- [ ] Read the student instructions and the synopsis PPT template on sfic.co.in
- [ ] By 3 Nov: breadboard site node printing temperature, humidity and globe temperature; FastAPI skeleton with a /readings endpoint
- [ ] By 6 Nov: synopsis PPT submitted, with a photo of the working node and one real WBGT reading

Risks and open questions

Risk	What we do about it
Cheap sensors read wrong	Check once against a reference (IMD station, or a calibrated meter borrowed from a college lab); pitch as early warning, not certified measurement
No pilot site in Kachchh	Ask the DDO's resource person for one site; fall back to a construction site or college workshop near Gandhinagar, labelled as such
Workers won't wear the badge	Vest-clip size; badges only for high-dust jobs; heat measured by the site node, not on the body
Workers hide results (fear of job loss)	Worker owns the record; employer and DDO see only anonymous totals
Only 4 days between shortlist and finale	Start building on 7 Nov, before shortlisting

Open questions:

- [ ] Can one team submit for more than one statement? (check the student instructions on sfic.co.in)
- [ ] Does the synopsis PPT template limit slides or sections?
- [ ] Who can connect us to a Kachchh site or the DDO's resource person?
- [ ] Can NIOH Ahmedabad or a factory medical officer review our check-up rules?
- [ ] Do the Gujarat OHS Audit Rules require medical-surveillance records?

Sources

- Seva First Innovation Challenge – timeline and problem statements
- Section 41C, Factories Act 1948 (AdvocateKhoj)
- OSH Code enforced from 21 Nov 2025 (Taxmann)
- OSH Code Third Schedule – notifiable diseases (TaxClue)
- Gujarat Factories (OHS Audit) Rules 2025 (Lexplosion)
- India's 2024 heat labour losses, Lancet Countdown 2025 (Outlook Business)
- Heat stress in vulnerable workers, Kidney Int Rep (PMC)
- Difficulties in recording occupational diseases (ILO)
- Gujarat's silent epidemic: silicosis (Down To Earth)
- OccuCare (Capterra India)
- Jarsh Safety (YourStory)
- Proxgy smart helmets (NITI Frontier Tech)
- Blackline Safety G7c
- Our JEM starter workbook (PS000109_JEM_starter.xlsx): job list, exposure scores and evidence